TwinForge: Toy Box Tactics

Situation report, Wednesday October 14, 2026, 9:00 PM

Bottom line

The game is playable end to end tonight with all eight toys as real 3D models, on the iPhone and on the laptop. Live at mattrheadland.github.io/twinforge (reload once to pick up the new build). 65 unit tests and 89 end-to-end tests pass. Two full-campaign runs through the real UI passed on the iPhone layout. The one thing I could not do is the live playtest in your Chrome, because the Chrome window stayed hidden all evening and Chrome throttles hidden tabs to a crawl.

Where the roster stands

Toy	Model	Joints / notes
Battle Rex	469 KB	head, jaw, arms, tail; swappable eyes
Spooky Tree	365 KB	branch arms; carved face stays painted
Ragin' Rockfish	559 KB	head, jaw, arms, tail, side eyes; small arm swings (forearms fused to hips in the mesh)
Young Arthur	443 KB	head, arms; swappable eyes
Wanda Wickwhistle	453 KB	REDESIGNED tonight: tall wicked sorceress. Head (hair tower + hat), broom arm, spark arm
Czernobog	428 KB	REDESIGNED tonight: giant obsidian demon god, horns, tattered wings. Head, arms, wings sway
Remy Roo	397 KB	head, front legs, tail; eye decal. Second Hunyuan job (first had sides flipped)
Billy Bass	429 KB	jaw, head, pectoral fins, tail; turned side-on so he reads as a fish

Czernobog's crew (Shadow Imp, Jitterbones) were rebuilt as lanky dancing silhouettes after your mountaintop reference. All designs are our own; the Disney references were used for mood only.

Fixed tonight

- Hammering the power button answered the learning question at random (the button sits under the right-hand answer). Taps are now ignored until the answers have popped in and the tap burst pauses.
- Rolling the die and tapping FIGHT together could start two battles. One at a time now.
- Board music was silent after the first victory jingle. Fixed.
- Trophy screen overflowed the screen; Next/Home/Go buttons had no click sound. Fixed.
- Arcade wiped an unfinished Tournament (shared save slot). Each mode has its own run now, with save migration.
- Reset deleted the backup and then copied old progress straight back; restoring from backup could be overwritten by a corrupt main save. Both fixed with tests.
- A battle step that throws now recovers instead of freezing the controls; Pause then Home always works.
- After a GPU reset (laptop sleep/wake) the campaign board came back black. The scene rebuilds now.
- Laptop: the "turn your phone" overlay no longer blocks a tall desktop window; select, board, HUD, question card and results fit from 4:3 to 21:9; the select screen no longer stutters (portraits cached); keyboard works (Enter/Space, arrows, Escape).

• Word banks moved out of the bundle (public/words.json): index.html is 975 KB against a 1 MB budget.

How long a campaign takes

Measured on a full automated run: 15 battles (13 creature fights and bosses on the path, plus retries), 61 turns, 48 learning questions, 18 die rolls, and 4.5 minutes of pure animation. Adding a 4-year-old choosing moves (about 8 s each) and answering questions (about 12 s each):

	Per world	Whole campaign
Fast, few wrong answers	4-6 min	25 min
Typical	6-9 min	35 min
Slow, several retries	9-12 min	45-50 min

It saves between fights, so this is really 2-3 sittings for Luca.

Learning content

22 question types (13 math, 9 reading) across 5 self-adjusting levels; three right in a row moves him up. Reading starts at letter matching, then starting sounds, rhymes, missing letters, read-a-word, ending sounds, sight words, and spelling at level 5. Pool: about 120 picture words, 26 rhyme families, 41 words to read, 44 sight words; hundreds of distinct questions per subject from level 2 up. The last 25 questions never repeat and a missed skill comes back 3-5 questions later in a new form. Every question is read aloud.

Risks and what is not verified

• No live playtest in your Chrome tonight (window hidden). Earlier today the title screen held a flat 60 fps on the laptop; the rest is verified only in headless Chromium at 874x402 and 1280x720.
• Read-aloud uses the device voice and cannot be tested headless. Needs a tap on the iPhone first.
• Offline right after an update drops to code-built toys until the models re-cache (service worker only pre-caches the page).
• Bundle is 25 KB under the 1 MB budget; the next big feature needs to move something else out.
• Bog Shack floor texture from Astra is sitting in the ChatGPT TwinForge project, not yet in the repo (the arena borrows the pier planks).
• Two campaign runs at desktop size failed, but only while the test machine was overloaded (2-7 fps); they passed on the iPhone layout and the same steps pass in e2e. Worth one more run on a quiet box.

Process note

The helper agents I tried to launch were declined by a permission prompt on your side every time, which cost about two hours. Everything after 7:40 PM was done solo. If you want parallel helpers next time, approve the "launch agent in a worktree" prompt when it pops up in the app, or tell me up front to stay solo.

Next steps

• Play it with Luca; send the learning report ("Copy as text") plus anything too hard or confusing.
• With Chrome in front on the laptop: 15-minute live playtest and frame-time check.
• Drop the Bog Shack floor into the repo and run scripts/build-arenas.py.
• iPhone pass: voice reads questions, emoji pictures read right, add to home screen.